

(he/him)

PhD in Economics | Data, Causal Inference & Machine Learning

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 JulianStreyczek

 [julian-streyczek](#)

Personal Profile

I bring 5 years of experience owning end-to-end analytical and ML projects that translate complex questions about user behavior on digital platforms into robust causal evidence and predictive models. Skilled in causal inference, machine learning, NLP (text embeddings), LLM-based data collection, human evaluation, metric design, and reproducible pipelines (Python, R, SQL, Git). Deep familiarity structuring ambiguous problems, managing projects and stakeholders, mentoring, and presenting results. Clear communication, pragmatic execution, and the ability to adapt and grow quickly.

Experience

- **Data Science Projects:** Led the full research cycle on complex projects:
 - Designed a quasi-experimental analysis (*Staggered Diff-in-Diff*) on web traffic data of news websites to quantify the impact of subscriptions (*paywalls*) on consumer behavior, showing average declines in website views of 25–30%
 - Developed a novel quasi-experimental analysis on panel data of scientists by combining ML and causal inference methods (*IV*, *XGBoost*, *Ridge*, *text embeddings*) to show nuanced effects of social media usage on productivity and networks
 - Trained predictive models (*Logistic Regression*, *XGBoost*) on user-level panel data to estimate individual propensity of subscribing to news websites and signing up to social media platforms
 - Built and maintained a large-scale dataset on scientists spanning 80k individuals, 2m publications, 80m citations, and 1000+ features by integrating SQL databases, APIs, web scraping, NLP pipelines (*embeddings*), and high-performance computing
 - Designed an LLM-based (*OpenAI Batch API*) data collection and human evaluation pipeline to collect and validate educational history and social media accounts of 80k individuals from web sources, reducing cost and time investment by >90%
 - Coordinated 10 research assistants over 2 years for data labeling; worked in geographically dispersed research teams of up to 5
- **Teaching Assistant, Bocconi University, Italy:** Designed and delivered theory, statistics, and ML lectures to mixed technical and non-technical audiences; mentored students on analytical projects (*Undergraduate & Graduate*) 2021 – 2026
- **Research Assistant, ZEW – Leibniz Institute, Germany:** Analyzed competitive dynamics in digital platforms; produced visualizations and analytical summaries for internal decisions 2018 – 2019
- **Consulting Intern, EY Innovalue, Germany:** Worked with international payment technology provider to deliver market analyses and strategic recommendations as executive slide decks 2018

Education

PhD Economics, Bocconi University, Italy

Sep 2020 – Jan 2026

- **Specialization:** Applied Causal Inference, Machine Learning, User Behavior
- **Awards:** German Academic Scholarship Foundation, Bocconi PhD Scholarship, Invernizzi Scholarships 2024, 2025 (€100,000+)
- **Grade:** cum laude (highest honors)

Visiting Fellow, *Harvard University, United States*

Jan – May 2025

- Invited researcher sponsored by Prof. Jim Snyder; presented ongoing work in seminars and engaged with faculty

MSc & BSc Economics, Mannheim University, Germany

2014 – 2020

- **Specialization:** Theoretical & Applied Statistics, Machine Learning, Digital Platforms
- **MSc Grade:** 1.0 / 5.0 (1.0 is best), with distinction
- **BSc Grade:** 1.4 / 5.0 (1.0 is best), top 10%

Skills

Coding	Python (pandas, numpy, scipy, statsmodels, scikit-learn, xgboost, lightgbm, matplotlib, seaborn, plotly), SQL (BigQuery, SQLite), R (tidyverse, dplyr, fixest, mlr3, ggplot2)
Data	Panel data, time series, geospatial data, databases, NLP (text), embeddings, LLM APIs, human evaluation
Causal Inference	Experiment design, hypothesis testing, double/debiased ML, causal forests, difference-in-differences, synthetic controls, regression discontinuity, instrumental variables
Machine Learning	Regression, classification (Logistic Regression, Random Forests), penalization (Lasso, Ridge, Elastic Net), gradient boosting (XGBoost, LightGBM), feature engineering, cross-validation, text embeddings
Tools & Engineering	Git, GitHub, Docker, PyCharm, BigQuery, AWS, GCP, Claude Code, Codex
Languages	German (native), English (fluent), Italian (basic)

Activities

Volunteer Student Mentor at Boarding School Schloss Hansenberg

since 2024

Violinist & Social Events Organizer at University Orchestra Mannheim

2015 – 2019

Member & Elected Representative in Economics Student Union Mannheim

2014 – 2019

Certified skiing instructor in Tyrol, Austria

2015 – 2017